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Mini Project report

On

“Blockchain-Based Land Registry System”

Submitted in partial fulfillment for Sixth semester of B.Tech (Regular) in

COMPUTER SCIENCE AND ENGINEERING

(IoT, Cyber Security Including Block Chain Technology)

By

Ramsha Siddiqui	**202256108035**
Durga Rajput	**202456108707**

Under the guidance of

Ms. Bhagat C. B.

(Asst. Professor)

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

CERTIFICATE

This is to certify that **Ramsha Siddiqui** (PRN: **202256108035**) and **Durga Rajput** (PRN: **202456108707**) have successfully completed their Mini-Project entitled **“Blockchain-Based Land Registry System”** in the Course of **“Blockchain”** in the Sixth Semester during tenure of completing the B.Tech program in Computer Science and Engineering (IoT, Cyber Security Including Blockchain Technology) from School of Engineering and Technology, MGM University institute.

Ms. Bhagat C. B.
Guide
CSE Department

Dr. Madhuri Kawarkhe
HOD
CSE Department

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1. Abstract

Land registration is an essential pillar of modern economic administration, establishing authoritative legal title over real property assets. However, conventional land records rely heavily on centralized departmental databases and physical paper registries, which remain inherently vulnerable to record tampering, fraudulent deed duplicate issuance, unauthorized alterations, and bureaucratic opacity. In developing economies, land disputes account for a significant share of all civil litigation due to unreliable chain-of-title archives.

To resolve these vulnerabilities, this project develops **TerraChain**, a production-grade, decentralized **Blockchain-Based Land Registry and Title Transfer System** deployed on EVM-compatible blockchain infrastructure. Rather than relying on rudimentary client-side simulations, the system is architected using **Solidity (v0.8.20)** smart contracts, the **Hardhat** development framework, **Pinata IPFS (InterPlanetary File System)** decentralized file storage, **Ethers.js (v6)**, and a reactive **React 18 (Vite + Tailwind CSS)** Web3 user interface.

All parcel registrations, surveyor blueprints, and deed PDFs are cryptographically anchored to IPFS, permanently storing the immutable 32-byte Content Identifier (CID) directly within the smart contract. A two-tier gatekeeper workflow requires designated Government Land Inspectors to physically audit deeds and verify titles on-chain before a property can enter the marketplace. Once verified, owners can list parcels, and prospective buyers execute trustless, atomic purchases governed by smart contract escrow with OpenZeppelin ReentrancyGuard, ensuring simultaneous ETH payment settlement, title reassignment, and automatic surplus payment refunds. The smart contract integrity is validated through an automated test suite of 21 comprehensive test cases with 100% assertions satisfied.

2. Introduction

Traditional land administration frameworks suffer from severe systemic inefficiencies. In centralized relational database architectures (such as standard SQL systems), any compromised database administrator credential, rogue employee, or system vulnerability allows direct UPDATE or DELETE operations on property ownership records without producing an undeniable cryptographic audit trail. Furthermore, prospective property buyers face immense friction, high legal search fees, and weeks of delay when attempting to verify historical chains of title across fragmented municipal archives.

Blockchain technology eliminates the single point of failure by introducing a distributed, cryptographically secured, append-only state machine. In Ethereum and EVM-compatible networks, state transitions are governed by self-executing smart contracts whose bytecode is immutably deployed to the ledger. Once written, a property title cannot be altered, forged, or backdated by any external entity or single official.

However, deploying real estate systems on public blockchains presents a technical bottleneck: storing multi-megabyte deed PDFs, satellite GIS surveys, and cadastral boundary maps directly on-chain is cost-prohibitive due to gas fees. TerraChain resolves this challenge through a **hybrid on-chain/off-chain decentralized architecture**: high-resolution deed documents and survey plans are stored in decentralized peer-to-peer storage via **Pinata IPFS**, while the cryptographic Content Identifier (CID), ownership states, verification flags, and atomic escrow logic are executed on-chain via **Solidity smart contracts**. This design delivers mathematical proof of title provenance, eliminates predatory middlemen, enforces role-based government verification, and provides instantaneous settlement for property transactions.

3. Course Outcomes Integrated

List of Course Outcomes of Course which are Related to Mini Project:

- **CO-1:** Understand and apply cryptographic hashing functions (Keccak-256 and SHA-256) alongside decentralized content-addressing (IPFS CID) to ensure tamper-evident storage of legal documents and instantaneous detection of data corruption.

• **CO-2:** Analyze distributed ledger architectures, EVM state transitions, gas optimization techniques, and consensus mechanisms (Proof-of-Stake / Proof-of-Authority) in enterprise and public blockchains.

• **CO-3:** Design and develop responsive Web3 user interfaces utilizing React 18, Tailwind CSS, Vite, and Ethers.js v6 with MetaMask provider/signer integration (EIP-1193).

• **CO-4:** Model secure transactional smart contracts in Solidity (v0.8.20), role-based access control (Admin / Land Inspector, Registered Owner, Buyer), and atomic escrow title transfer mechanisms.

• **CO-5:** Evaluate cybersecurity threats in decentralized smart contracts (reentrancy attacks, integer overflow, unauthorized access) and demonstrate cryptographic resilience using OpenZeppelin ReentrancyGuard, custom errors, and automated unit testing (21/21 passing tests).

4. Action Plan

S. No.	Details of activity	Planned Start	Planned Finish	Team Members
1	Problem Formulation, Literature Survey & EVM/IPFS System Architecture Formulation	Week 1	Week 2	Ramsha Siddiqui & Durga Rajput
2	Smart Contract Engineering: Solidity 0.8.20 LandRegistry, ReentrancyGuard, Custom Errors & Escrow	Week 2	Week 3	Ramsha Siddiqui
3	Automated Testing Suite: Hardhat Unit Tests (21/21 Tests Passed), Edge Cases & Balance Assertions	Week 3	Week 4	Durga Rajput
4	Decentralized Storage: Pinata IPFS Service Integration for Deed PDF Pinning & Gateway Resolution	Week 4	Week 5	Ramsha Siddiqui
5	Web3 Frontend Development: React 18, Vite, Tailwind CSS, Ethers.js v6 & MetaMask Role Dashboard	Week 5	Week 6	Durga Rajput
6	Comprehensive System Verification, Localhost Node Deployment, Cross-Role Flow Testing & Documentation	Week 6	Week 7	Ramsha Siddiqui & Durga Rajput

5. Literature Review

Property registration systems globally are undergoing a digital transformation. The earliest land registration systems were manual, deed-based registries wherein written paper documents were cataloged in government record offices. While these paper registries served their historical purpose, they introduced severe vulnerabilities including physical deterioration, loss from natural disasters, and illegal forgery.

During the late 20th and early 21st centuries, governments transitioned towards computerized centralized databases. While computerized land systems improved indexing and search speed, they failed to resolve fundamental cybersecurity and governance issues. In a standard relational database (e.g., MySQL or Oracle), any user possessing root administrative privileges can execute direct UPDATE or DELETE queries on ownership rows. As there is no cryptographic linkage connecting subsequent entries, fraudulent alterations can remain completely undetected for decades.

The emergence of blockchain technology—formalized by Satoshi Nakamoto in 2008 and expanded to Turing-complete smart contracts by Ethereum in 2015—introduced an unalterable data structure termed a cryptographic distributed ledger. Each state transition is executed deterministically by smart contracts and validated across network nodes. Internationally, national land registry authorities have tested blockchain integration: (1) **Republic of Georgia** partnered with Bitfury and NAPR in 2016 to anchor title hashes to a public blockchain network, processing over 1.5 million deeds with zero disputes; (2) **Sweden (Lantmäteriet)** tested smart contract-based land sales reducing closing times from months to hours; and (3) **India (Andhra Pradesh & Telangana)** initiated pilots in Amaravati and Hyderabad safeguarding agricultural land allotments.

Crucially, existing literature identifies that storing raw document files directly on-chain is economically infeasible. By utilizing IPFS (InterPlanetary File System) to anchor cryptographic Content Identifiers (CIDs) on-chain, TerraChain eliminates this storage bottleneck while maintaining absolute document authenticity.

6. Proposed Methodology

The proposed system implements a modular 3-tier decentralized architecture operating across smart contract execution, peer-to-peer storage, and modern Web3 client technologies:

1. Tier 1: Ethereum Smart Contract Layer (Execution Engine): Developed in Solidity 0.8.20 and compiled via Hardhat. The core LandRegistry.sol contract maintains the canonical state of all parcels on the EVM. The Land struct tracks: id, physicalAddress, areaSqFt, priceInWei, ipfsHash, currentOwner, isVerified, and isForSale. The contract enforces role-based access control using custom modifiers (onlyAdmin and onlyOwner), gas-optimized custom errors, and an atomic escrow buyLand() function incorporating OpenZeppelin ReentrancyGuard.

2. Tier 2: Decentralized Storage Layer (Pinata IPFS): Cadastral survey boundary maps, notarized sale deeds, and identity proofs are uploaded directly to IPFS nodes using Pinata's REST API. The system generates an immutable 32-byte Content Identifier (CID). Storing the CID in the smart contract guarantees that if even a single pixel or character of the deed PDF is modified, the CID changes, immediately exposing the tampering.

3. Tier 3: Client Presentation & Web3 Integration Layer: Built using React 18, Vite, and Tailwind CSS. Ethers.js v6 handles RPC communication with MetaMask (EIP-1193). The application automatically detects the connected wallet's persona (Admin / Land Inspector, Registered Owner, or Prospective Buyer) and dynamically renders role-specific dashboards with real-time transaction toast notifications.

7. Implementation

The application is organized as an enterprise-grade monorepo containing two dedicated workspaces:

- **blockchain/contracts/LandRegistry.sol:** Production smart contract exposing 7 core functions: registerLand (unverified parcel creation), verifyLand (Inspector title audit), listForSale (owner listing), delistLand (withdraw listing), buyLand (atomic escrow payment & title transfer), getLandDetails (metadata query), and getLandsByOwner (portfolio query). Emits events: LandRegistered, LandVerified, LandListed, LandDelisted, LandSold.
- **blockchain/scripts/deploy.js:** Hardhat deployment script that deploys to Localhost (Chain ID: 31337) or Sepolia testnet, automatically exporting contract address and ABI to frontend/src/contracts/LandRegistryConfig.json.
- **blockchain/test/LandRegistry.test.js:** Automated test suite comprising 21 unit tests with Chai assertions, validating deployment, input bounds, role restrictions, atomic balance settlements, and ReentrancyGuard.
- **frontend/src/services/pinataService.js & DocumentUploader.jsx:** Axios-based Pinata IPFS file ingestion component supporting drag-and-drop, 15MB size enforcement, live upload progress bars, and gateway URL generation.
- **frontend/src/context/Web3Context.jsx:** Ethers.js v6 state management binding to MetaMask (window.ethereum), handling account changes, chain switching, and contract instances.
- **frontend/src/App.jsx:** Main dApp dashboard providing 4 role-based views (Explore Marketplace, Register Property, Inspector Admin Portal, My Assets) and floating toast notifications.

8. Outputs of the Mini-Projects

The completed system executes all functional land registry workflows with real-time on-chain verification:

1. Wallet Connection & Multi-Persona Detection: MetaMask connects with the local Hardhat node (Chain ID: 31337). The UI automatically identifies whether the user is the Contract Admin (Inspector: 0xf39F...), an Existing Owner (0x7099...), or a Buyer (0x3C44...).

2. Land Parcel Registration with IPFS Pinning: A landowner inputs parcel specifications (Street Address, 2400 sq ft, 2.5 ETH) and drops a deed PDF into the DocumentUploader. The file is pinned to Pinata IPFS (CID: Qm...), and registerLand() executes on-chain with isVerified = false.

3. Government Inspector Title Verification: The Admin switches to the Inspector Panel, inspects the uploaded IPFS deed via gateway, and calls verifyLand(). The status updates to a verified green badge.

4. Marketplace Listing: The verified owner accesses 'My Assets', inputs an asking price in ETH, and invokes listForSale(), publishing the parcel to the public marketplace.

5. Atomic Escrow Purchase & Title Transfer: A buyer clicks 'Buy Now' with 2.5 ETH. The smart contract atomically transfers 2.5 ETH to the seller, reassigns ownership to the buyer, resets listing status, and refunds any excess payment atomically.

6. Automated Verification Matrix: All 21 Hardhat unit tests pass in 798ms, proving 100% contract security and functional correctness.

9. Skill Developed / learning out of this Mini-Project

• **Solidity Smart Contract Mastery:** Designed robust EVM smart contracts utilizing Solidity 0.8.20, state storage variables, custom data structures, and gas-efficient custom errors.

• **EVM Security Patterns:** Implemented OpenZeppelin ReentrancyGuard, enforced the Checks-Effects-Interactions pattern, and implemented strict role-based access modifiers.

• **Hardhat Testing & Toolchain:** Built comprehensive automated test suites using Mocha and Chai, automated contract deployments with scripted ABI synchronization, and simulated multi-account local nodes.

• **Decentralized Storage Integration:** Integrated IPFS and Pinata REST APIs for off-chain content-addressed file storage and dynamic CID resolution.

• **Web3 Frontend Engineering:** Developed reactive frontends using React 18, Vite, and Tailwind CSS, leveraging Ethers.js v6 for wallet communication, BigInt arithmetic, and EIP-1193 event listeners.

• **Technical Teamwork & Academic Documentation:** Acquired practical experience in full-stack dApp lifecycle management, software architecture, and academic engineering documentation.

10. Applications of this Mini-Project

1. Municipal and State Land Revenue Registries: Government land registries can replace vulnerable relational SQL databases with an immutable smart contract ledger, completely eliminating forged deeds, illegal double-selling of plots, and collusive backdated changes by corrupt officials.

2. Transparent Agricultural Land Bank Records: Rural property borders and multi-owner family inheritances often suffer from protracted litigation. An immutable public ledger establishes a clear, undeniable line of historical title succession.

3. Commercial Bank Mortgage Due Diligence: Commercial banks and financial institutions can instantly inspect on-chain verified deeds and title histories, eliminating weeks of manual legal searches and preventing fraudulent multi-bank pledging.

4. Commercial Real Estate Leasing & Smart Escrow: Corporate developers and leasing agencies can record long-term leasehold deeds and execute high-value real estate purchases without expensive third-party escrow fees.

5. Cadastral Boundary & Surveyor Blueprint Tracking: Satellite GIS survey files and municipal master plans can be permanently pinned to IPFS, ensuring public transparency over civic zoning and land-use designations.

6. Academic & Research Laboratory Tool: Serves as an interactive, fully functional educational platform for computer science and IT students to inspect how smart contracts, IPFS content addressing, and atomic escrow operate in practice.

11. Area of Future Improvement

While the current implementation successfully models and validates all core blockchain principles with Solidity, Hardhat, IPFS, and React, several advanced extensions can be incorporated in future iterations for enterprise nationwide deployment:

1. **Multi-Signature Consensus for Government Registrars:** Implementing multi-signature (multi-sig) authorization requiring the simultaneous digital keys of the Property Buyer, Seller, Municipal Town Planner, and Sub-Registrar Officer before confirming a title transfer.
2. **Government Biometric KYC & Digital Identity (DID):** Connecting land transfer approvals to national digital identity systems (such as Aadhaar e-KYC and digital signatures) to cryptographically verify buyer and seller identity prior to transaction execution.
3. **Zero-Knowledge Proofs (zk-SNARKs) for Financial Privacy:** Integrating zk-SNARKs to allow property buyers and sellers to verify transaction solvency and legal compliance without broadcasting private sale sums to public blockchain explorers.
4. **Layer-2 Rollups (Arbitrum / Polygon) Deployment:** Migrating contract deployment from Ethereum mainnet to Layer-2 optimistic or zero-knowledge rollups to achieve sub-cent transaction gas fees and high transaction throughput.
5. **Decentralized Legal Dispute Arbitration (Kleros Protocol):** Integrating decentralized crowd-jury arbitration protocols such as Kleros to resolve boundary disputes and inheritance contestations through economic staking incentives.

12. Conclusion

Land registration is a foundational pillar of modern governance, yet conventional manual and centralized digital database registers continue to suffer from corruption, single points of failure, unrecorded alterations, and rampant litigation. This mini-project successfully engineered, deployed, and verified **TerraChain**, a production-ready, decentralized blockchain-based land registry system.

By integrating **Solidity 0.8.20 smart contracts**, **Hardhat**, **Pinata IPFS decentralized deed storage**, **Ethers.js v6**, and a responsive **React 18 Web3 user interface**, the system establishes a permanent, auditable, and mathematically unalterable record of property ownership. The smart contract enforces a two-tier gatekeeper architecture where Government Land Inspectors authenticate cadastral documents before marketplace trading is enabled. Property purchases settle atomically with full reentrancy protection and automatic surplus refunds. Validated against 21 automated unit tests with 100% pass rates, TerraChain conclusively demonstrates how blockchain technology can eradicate title fraud, streamline land administration, and restore citizen trust in digital property governance.